

Nguyễn Đức Anh (Aldric)

Senior Full-Stack Engineer — TypeScript · Node.js · React · Web3

Backend-heavy full-stack engineer with deep Web3 experience. Building DeFi provider integrations, launchpads, and long-running on-chain flows on the stack LI.FI runs — Node.js/TypeScript, React + viem, ethers.js, MongoDB, Redis, Docker/Kubernetes on AWS.

Remote · anhnd350309@gmail.com · github.com/0xaldrich · linkedin.com/in/0xaldrich

Summary

Senior full-stack engineer with 5+ years shipping production backend systems and 3+ years building Web3 products end to end — DEX routing, launchpads, NFT execution engines, and cross-chain tooling. Comfortable across the core LI.FI stack: TypeScript/Node.js, React with viem, ethers.js, MongoDB/Redis, Docker/Kubernetes on AWS. Strong instincts around provider integrations, async flows, retries/idempotency, observability, and recovery — the exact failure modes a Checkout product has to absorb.

Experience

Senior Web3 Engineer (Freelance) · Independent

05/2025 — Present

Remote

Building production crypto products end to end for multiple clients — DeFi swap/bridge aggregation, token launchpads, and on-chain oracles — on TypeScript/Node.js, ethers + viem, and Solidity.

- BinkOS — a DeFi execution layer that normalizes 10+ swap, bridge, and lending providers (deBridge cross-chain, PancakeSwap, KyberSwap, OKX DEX, Jupiter, Venus) behind one provider-agnostic plugin interface — the same provider-normalization and routing problem a Checkout product solves. TypeScript monorepo on ethers + viem + @solana/web3.js.
- Token launchpad tooling — launch and buy/sell flows on Solana (LetsBonk) and EVM, with on-chain execution via ethers.js, IPFS metadata, and idempotent state — covering the funding-flow edge cases (slippage, retries, partial fills) Checkout has to absorb.
- On-chain oracle — real-time pipeline turning live data into commit-reveal attestations signed with eth-account and verified on-chain, with orphan/recovery handling for long-running, failure-prone rounds.
- Hands-on AMM internals — a DEX on TON (router, pools, LP accounting) plus EVM contracts in Foundry, giving working depth on routing and liquidity behaviour.

Stack — TypeScript, Node.js, ethers.js, viem, @solana/web3.js, Solidity, TON / FunC, MySQL, Redis, Docker

Open-source Contributor · Mojave · 1sixtech

04/2025 — 07/2025

Remote · Open Source

Bitcoin / zk infrastructure — Rust workspace for full-node, sequencer, prover, and cryptographic utilities.

- Implemented block signing and verification (PR #17) — cryptographic identity guarantees for produced blocks.
- Built automatic ethrex upgrade tooling (PR #20) — keeping the workspace in lockstep with upstream changes.
- Worked across a multi-crate Rust workspace covering full node, sequencer, prover, and cryptographic primitives.

Stack — Rust, Cryptography, ZK, Bitcoin, ethrex

Senior Software Engineer · Topology Foundation

12/2024 — 03/2025

Remote (distributed: Germany · France · Korea)

Decentralized Realtime Program (DRP) — open-source primitives for real-time collaborative apps. Worked daily with teammates across CET (Germany, France) and APAC (Korea); standups, pair-programming, and on-call all built around a shared European afternoon overlap.

- Improved build processes and CI/CD workflows for the DRP runtime, supporting modular code evolution.
- Authored high-coverage test suites — lifted code coverage to 90% across distributed components.
- Integrated performance metrics for network latency and CPU time, enabling real-time profiling and observability in decentralized environments.
- Operated effectively across a CET/APAC team split — proof point for any EMEA-leaning role considering an APAC contributor.

Stack — TypeScript, Node.js, Vitest, GitHub Actions, Distributed Systems

Head of Engineering · AnyAxis Labs

02/2024 — 12/2024

Remote

Led the engineering org behind multiple Web3 products across the TON ecosystem — DEX, launchpad, and token issuance.

- AytuDEX — engineered algorithmic routing optimizations 50–100% faster than peer DEXes on the same pairs (directly applicable to bridge / DEX-aggregator routing logic).
- TonPad — designed launchpad smart contracts and system architecture: allocation, vesting, treasury, partner-configurable rules.
- Pump.fun-style platform — separated read / write / socket layers for predictable scaling under bursty long-running flows.
- Led a high-availability backend supporting 50,000+ active users at 99.9% uptime; standardized deploys, observability, and incident response on K8s.
- Built and directed a backend + blockchain team; owned hiring, technical roadmaps, and security posture.

Stack — TypeScript, NestJS, Solidity, TON / FunC, PostgreSQL, Redis, Kafka, Kubernetes

Software Engineer (Part-time) · WispSwap Lab

03/2023 — 01/2024

Remote

Off-chain infrastructure for a DeFi DEX — the same kind of provider/indexer layer that sits behind a Checkout product.

- Designed and shipped the off-chain indexing + quote service for WispSwap; operated services for ~20,000 users and 1,000+ concurrent on peak days.
- Built idempotent quote refresh, retry/backoff, and reconciliation against on-chain truth.

Stack — TypeScript, Node.js, PostgreSQL, Redis, ethers.js

Software Engineer L3 · Teko

12/2021 — 06/2023

Enterprise CRM and merchant tooling for a major retail tech group — production backend foundations.

- Built enterprise CRM apps on FastAPI, Elasticsearch, Redis, MySQL, and Kafka — cut API latency by 60% via targeted caching.
- Maintained legacy systems while upgrading with SOLID + design patterns; kept test coverage above 95%.
- Mentored four new team members through code reviews and on-the-job guidance.

Stack — Python, FastAPI, MySQL, Redis, Elasticsearch, Kafka

Skills

Languages	TypeScript, Solidity, Rust, Python, Go (learning), SQL
Backend	Node.js, NestJS, Express, FastAPI, microservices, REST, WebSockets
Web3 / on-chain	ethers.js, viem, @solana/web3.js, Foundry, Hardhat, EVM, Solidity, TON / FunC, Solana
Frontend	React, TypeScript, Tailwind, viem for Web3 UI (wagmi familiar)
Data & messaging	MongoDB, PostgreSQL, MySQL, Redis, Elasticsearch, Kafka, RabbitMQ
Cloud & DevOps	Docker, Kubernetes, AWS, CloudFlare, GitHub Actions, GitOps CD, OpenTelemetry, Prometheus / Grafana
Integration patterns	Provider normalization, idempotent quotes, async retries, webhooks, reconciliation, recovery flows, circuit breakers

Selected projects

BinkOS — DeFi provider-aggregation layer (freelance) — Normalizes 10+ swap / bridge / lending providers (deBridge, PancakeSwap, KyberSwap, OKX DEX, Jupiter, Venus) behind one plugin interface — the exact provider-normalization + routing thesis LI.FI is built on. TypeScript monorepo on ethers + viem + @solana/web3.js.

Token launchpad tooling (freelance) — Launch and buy/sell flows on Solana (LetsBonk) and EVM — on-chain execution via ethers.js, IPFS metadata, idempotent state. Handles slippage, retries, and partial fills.

On-chain oracle (freelance) — Real-time pipeline producing commit-reveal attestations signed with eth-account and verified on-chain, with orphan/recovery handling for long-running rounds.

AytuDEX / TonPad / Pump.fun-style platform — Web3 systems shipped via AnyAxis Labs: custom DEX routing (50–100% faster), launchpad smart contracts, scalable token issuance with separated read/write/socket layers.

Mojave · Bitcoin / zk infrastructure — Rust workspace for full-node, sequencer, prover, and cryptographic utilities.

Contributed block signing & verification (PR #17) and automatic ethrex upgrade tooling (PR #20). github.com/1sixtech/mojave

Education

Hanoi University of Science and Technology (HUST) — B.Sc., Computer Science · 09/2017 — 09/2022

Achievements

- UAVS Hackatrix — Top 7, "Application for job management and salary payment" (07/2021)
- KPMG Challenge — solutions to help businesses overcome pandemic impact
- HackerRank — Problem-Solving Certificate